

IMAP Client with TLS Support - Manual

Author: xfedor14 Tatiana Fedorova

Project Overview:

This C program (imapcl) implements a client for reading and downloading emails from an IMAP server using the IMAP protocol (RFC 5322).

The program allows for encrypted connections using SSL/TLS and supports various optional features such as downloading only new messages, fetching headers, and specifying an output directory for saving messages.

Key Features:

- Email Downloading: The program retrieves messages from the server and saves them as individual files in a specified directory.
- SSL/TLS Support: Optionally encrypts communication using SSL (port 993).
- Server Interaction: Connects to an IMAP server and allows interaction for email retrieval.
- Configurable: Accepts several command-line parameters for customization (server, port, authentication, output directory, etc.).

Command-Line Parameters:

```
./imapcl server [-p port] [-T [-c certfile] [-C certaddr]] [-n] [-h] -a auth_file [-b MAILBOX] -o out_dir
```

- server: The server's IP address or domain name (required).
- -p port: Specify the port for the server (optional; default 143 or 993 depending on encryption).
- -T: Enable TLS encryption (use IMAPS).
- -c certfile: File containing SSL/TLS certificates for server verification (optional).

- -C certaddr: Directory to search for certificates (optional, default /etc/ssl/certs).
- -n: Fetch only new (unseen) messages.
- -h: Fetch only message headers.
- -a auth_file: Path to authentication file containing the username and password.
- -b MAILBOX: Specify mailbox (default is "INBOX").
- -o out_dir: Output directory to save the downloaded emails.

Example Usage:

- Download emails from the server:

```
$ imapcl eva.fit.vutbr.cz -o maildir -a auth_file.txt
```

- Download only new emails with TLS encryption:

```
$ imapcl 10.10.10.1 -p 9993 -T -n -b Important -o maildir -a cred
```

- Handle server identity verification failure:

```
$ imapcl eva.fit.vutbr.cz -o maildir -a /dev/null
```

Authentication File Format:

The authentication file should contain the following format:

```
username = your_username
```

```
password = your_password
```

Program Flow:

1. Initialization: The program initializes necessary variables and structures using the init() function.
2. Argument Parsing: The program parses command-line arguments to set configuration values.
3. Connection Setup: The program establishes a connection to the IMAP server, optionally using SSL/TLS.
4. Login and Mailbox Selection: The program logs in using credentials and selects the specified mailbox (default: INBOX).

5. Message Retrieval: The program fetches message headers or bodies based on user flags and stores them in the specified output directory.
6. Error Handling: The program includes robust error handling, especially for network or allocation failures.

Notable Functions:

- help(): Displays program usage and options.
- destroy(): Frees allocated memory and resets program state.
- process_error(): Handles and reports errors, ensuring proper cleanup.
- init(): Initializes the program variables and structure.
- parse_program_arguments(): Parses command-line arguments and configures the program.
- process_auth_file(): Loads authentication credentials from the specified file.
- connect_common(): Establishes a connection to the server.
- login_and_select(): Logs in to the server and selects the desired mailbox.
- read_response(): Handles responses from the server, extracting relevant information such as message headers or body content.